SQL Data Analysis Report

E-Commerce Orders Dataset • 1,200 Records • 14 Columns

1,200	$1,264,761.96	$1,053.97	$3,456.40
Total Orders	Total Revenue	Avg Order Value	Max Single Order

1. Introduction

This report presents the findings of Project 3 of the DecodeLabs Data Analytics Industrial Training Program. The objective of this project is to demonstrate proficiency in SQL — the foundational language of data extraction and business intelligence — by writing structured queries against a real-world e-commerce dataset.

SQL (Structured Query Language) is a declarative language used to communicate with relational databases. Unlike procedural languages that describe how to retrieve data step by step, SQL allows the analyst to declare what result is needed, and the database engine determines the most efficient retrieval path.

The ten queries documented in this report collectively cover the full spectrum of fundamental SQL operations required for professional data analysis:

- Basic data retrieval using SELECT
- Row-level filtering using WHERE with text and numeric conditions
- Categorical grouping using GROUP BY
- Statistical aggregation using COUNT, SUM, and AVG
- Post-aggregation filtering using HAVING
- Combined filtering and grouping for multi-condition analysis
- Result ordering using ORDER BY

2. Dataset Overview

The dataset used for this analysis is a cleaned e-commerce orders table containing 1,200 records across 14 columns. Each row represents a single customer order. The table below summarizes the columns and their data types.

Column Name	Data Type	Description
OrderID	VARCHAR	Unique identifier for each order (e.g. ORD200000)
Date	DATE	Date the order was placed
CustomerID	VARCHAR	Unique identifier for the customer
Product	VARCHAR	Product category (Chair, Desk, Laptop, Monitor, Phone, Printer, Tablet)
Quantity	INT	Number of units ordered
UnitPrice	FLOAT	Price per unit in USD

Column Name	Data Type	Description
ShippingAddress	VARCHAR	Customer delivery address
PaymentMethod	VARCHAR	Payment type used (Cash, Credit Card, Debit Card, Gift Card, Online)
OrderStatus	VARCHAR	Current status (Cancelled, Delivered, Pending, Returned, Shipped)
TrackingNumber	VARCHAR	Shipment tracking code
ItemsInCart	INT	Total items in the cart at time of order
CouponCode	VARCHAR	Discount coupon applied (FREESHIP, NO_COUPON, SAVE10, WINTER15)
ReferralSource	VARCHAR	Marketing channel that drove the order (Email, Facebook, Google, Instagram, Referral)
TotalPrice	FLOAT	Final order value in USD

3. SQL Execution Order & Key Concepts

A critical insight for writing correct SQL is understanding that the logical execution order of a query differs from its written syntax order. The database engine processes clauses in the following sequence:

Step	Clause	Purpose
1	FROM / JOIN	Identifies the data source table(s)
2	WHERE	Filters individual rows before grouping
3	GROUP BY	Collapses rows into categorical buckets
4	HAVING	Filters aggregated groups (after GROUP BY)
5	SELECT	Picks columns and computes calculated fields
6	ORDER BY	Sorts the final output result set

Understanding this order prevents the common "Alias Trap" error — attempting to reference a SELECT alias inside a WHERE clause fails because WHERE is evaluated before SELECT has created the alias.

4. Query Analysis & Results

Query 1 — Basic SELECT: View All Orders

Objective: Retrieve all columns from the orders table to inspect raw data structure.

```
SELECT * FROM orders LIMIT 50;
```

Insight: *This foundational query confirms the table structure — 14 columns covering order identity, product, pricing, logistics, and marketing attribution. It is always the first step in any exploratory analysis.*

Query 2 — WHERE (Text Filter): Delivered Orders

Objective: Isolate only orders with a status of "Delivered", sorted by highest revenue first.

```
SELECT OrderID, CustomerID, Product, TotalPrice, OrderStatus FROM orders WHERE
OrderStatus = 'Delivered' ORDER BY TotalPrice DESC;
```

Insight: *231 orders have been successfully delivered, generating a combined revenue of $242,600.32. The highest single delivered order reached $3,456.40.*

Query 3 — WHERE (Numeric Filter): High-Value Orders Above $2,000

Objective: Extract premium orders where TotalPrice exceeds $2,000 to identify top-tier transactions.

```
SELECT OrderID, CustomerID, Product, Quantity, TotalPrice FROM orders WHERE TotalPrice >
2000 ORDER BY TotalPrice DESC;
```

Insight: *High-value orders (>$2,000) are predominantly quantity-5 orders, confirming that bulk purchasing is the primary driver of premium revenue. Chair and Tablet products appear frequently in this segment.*

Query 4 — GROUP BY + Aggregations: Sales Summary by Product

Objective: Aggregate total orders, quantity, revenue, and average order value grouped by product category.

```
SELECT Product,        COUNT(*) AS Total_Orders,        SUM(Quantity) AS Total_Quantity,
ROUND(SUM(TotalPrice), 2) AS Total_Revenue,        ROUND(AVG(TotalPrice), 2) AS
Avg_Order_Value FROM orders GROUP BY Product ORDER BY Total_Revenue DESC;
```

Product	Total Orders	Total Quantity	Total Revenue ($)	Avg Order Value ($)
Chair	178	562	195,620.11	1,098.99
Printer	181	542	195,612.61	1,080.73
Laptop	173	535	192,126.56	1,110.56
Tablet	179	497	186,568.95	1,042.28
Monitor	163	480	175,651.41	1,077.62
Desk	170	508	167,459.93	985.06
Phone	156	411	151,722.39	972.58

Insight: *Chair and Printer are virtually tied for the top revenue position (~$195K each). Laptop records the highest average order value at $1,110.56, indicating customers spend more per transaction. Phone has the lowest volume and revenue, suggesting either lower demand or a smaller price point.*

Query 5 — GROUP BY: Orders & Revenue by Payment Method

Objective: Determine which payment methods are most popular and which generate the most revenue.

```
SELECT PaymentMethod,        COUNT(*) AS Total_Orders,        ROUND(SUM(TotalPrice), 2) AS Total_Revenue,        ROUND(AVG(TotalPrice), 2) AS Avg_Order_Value FROM orders GROUP BY PaymentMethod ORDER BY Total_Orders DESC;
```

Payment Method	Total Orders	Total Revenue ($)	Avg Order Value ($)
Online	258	262,442.94	1,017.22
Cash	246	259,786.29	1,056.04
Credit Card	234	263,847.63	1,127.55
Debit Card	232	232,361.18	1,001.56
Gift Card	230	246,323.92	1,070.97

Insight: *Online payment leads in order volume (258 orders), but Credit Card generates the highest average order value ($1,127.55), suggesting credit card users tend to make larger purchases. Payment method distribution is balanced, indicating the platform successfully accommodates diverse customer preferences.*

Query 6 — GROUP BY: Order Status Distribution

Objective: Understand how orders are distributed across the five status categories.

```
SELECT OrderStatus,        COUNT(*) AS Total_Orders,        ROUND(SUM(TotalPrice), 2) AS Total_Revenue FROM orders GROUP BY OrderStatus ORDER BY Total_Orders DESC;
```

Order Status	Total Orders	Total Revenue ($)
Cancelled	250	276,396.21
Returned	247	243,277.70
Pending	237	256,328.15
Shipped	235	246,159.58
Delivered	231	242,600.32

Insight: *Only 19.25% of orders are successfully delivered, while 20.8% are cancelled — the highest category. This is a significant business concern. Combined, cancelled and returned orders represent 41.4% of all transactions, indicating a potential fulfillment or product quality issue requiring immediate investigation.*

Query 7 — GROUP BY: Marketing Channel Analysis

Objective: Evaluate the performance of each referral source to identify the most effective marketing channel.

```
SELECT ReferralSource,        COUNT(*) AS Total_Orders,        ROUND(SUM(TotalPrice), 2) AS Total_Revenue,        ROUND(AVG(TotalPrice), 2) AS Avg_Order_Value FROM orders GROUP BY ReferralSource ORDER BY Total_Revenue DESC;
```

Referral Source	Total Orders	Total Revenue ($)	Avg Order Value ($)
Instagram	259	275,285.45	1,062.88

Referral Source	Total Orders	Total Revenue ($)	Avg Order Value ($)
Email	250	261,808.55	1,047.23
Google	241	250,441.48	1,039.18
Facebook	228	250,410.90	1,098.29
Referral	222	226,815.58	1,021.69

Insight: *Instagram is the top-performing marketing channel, driving 259 orders and $275,285 in revenue. Facebook, despite having fewer orders than Email and Google, achieves the second-highest average order value ($1,098.29), indicating high-intent buyers. Marketing budget should prioritize Instagram and Facebook.*

Query 8 — GROUP BY: Coupon Usage Analysis

Objective: Measure the impact and adoption rate of each discount coupon code.

```
SELECT CouponCode,        COUNT(*) AS Times_Used,        ROUND(SUM(TotalPrice), 2) AS
Total_Revenue,        ROUND(AVG(TotalPrice), 2) AS Avg_Order_Value FROM orders GROUP BY
CouponCode ORDER BY Times_Used DESC;
```

Coupon Code	Times Used	Total Revenue ($)	Avg Order Value ($)
FREESHIP	313	335,036.99	1,070.41
NO_COUPON	309	322,401.41	1,043.37
WINTER15	292	302,483.54	1,035.90
SAVE10	286	304,840.02	1,065.87

Insight: *FREESHIP is the most-used coupon (313 uses) and drives the highest total revenue ($335K), suggesting free shipping is a strong purchase motivator. Notably, SAVE10 — a percentage discount — generates a higher average order value ($1,065.87) than WINTER15 ($1,035.90), indicating SAVE10 attracts higher-spending customers.*

Query 9 — HAVING: Products with Average Order Value Above $1,000

Objective: Use the HAVING clause to filter product groups where the average order value exceeds $1,000 — demonstrating post-aggregation filtering.

```
SELECT Product,        COUNT(*) AS Total_Orders,        ROUND(AVG(TotalPrice), 2) AS
Avg_Order_Value FROM orders GROUP BY Product HAVING AVG(TotalPrice) > 1000 ORDER BY
Avg_Order_Value DESC;
```

Product	Total Orders	Avg Order Value ($)
Laptop	173	1,110.56
Chair	178	1,098.99
Printer	181	1,080.73
Monitor	163	1,077.62

Product	Total Orders	Avg Order Value ($)
Tablet	179	1,042.28

Insight: *Five of seven product categories meet the $1,000 average threshold. Desk ($985.06) and Phone ($972.58) fall below the threshold, qualifying them as lower average-value categories. Laptop commands the premium position at $1,110.56 per order. This insight can inform premium product bundling strategies.*

Query 10 — WHERE + GROUP BY: Delivered Revenue by Product

Objective: Combine WHERE (row-level filter) and GROUP BY (aggregation) to analyze revenue from successfully delivered orders only — by product.

```
SELECT Product,        COUNT(*) AS Delivered_Count,        ROUND(SUM(TotalPrice), 2) AS
Revenue_From_Delivered FROM orders WHERE OrderStatus = 'Delivered' GROUP BY Product ORDER
BY Revenue_From_Delivered DESC;
```

Product	Delivered Count	Revenue from Delivered ($)
Laptop	40	40,714.43
Phone	38	40,345.41
Printer	29	38,054.73
Monitor	31	35,999.62
Tablet	28	31,794.52
Chair	33	31,465.83
Desk	32	24,225.78

Insight: *Laptop leads delivered revenue at $40,714.43 despite having the third-lowest delivered count (40), confirming it commands a higher per-unit price. Phone ranks second in delivered revenue ($40,345.41) with 38 deliveries, demonstrating strong per-order value. Desk has both the lowest delivered revenue and the smallest count of delivered orders.*

5. Key Business Insights

#	Area	Finding	Recommendation
1	Product Revenue	Chair & Printer lead total revenue (~$195K each)	Maintain inventory levels; consider bundle offers
2	Avg Order Value	Laptop has highest avg order value at $1,110.56	Promote upsells and accessories alongside Laptop
3	Order Fulfillment	41.4% of orders are Cancelled or Returned	Investigate root cause — logistics or product quality

#	Area	Finding	Recommendation
4	Marketing Channels	Instagram drives the most revenue ($275K)	Increase Instagram ad spend; replicate for Facebook
5	Payment Behavior	Credit Card users spend the most per order ($1,127)	Offer credit card-exclusive promotions
6	Coupon Strategy	FREESHIP is the most used and highest revenue coupon	Make FREESHIP the default promotional lever
7	Premium Segment	5 out of 7 products have avg order value above $1,000	Strong premium product portfolio — focus on retention

6. Conclusion

This project successfully demonstrates the application of fundamental SQL techniques to extract actionable business intelligence from a raw e-commerce dataset. The ten queries progressively build in complexity — from basic data retrieval with SELECT, through conditional filtering with WHERE, to categorical aggregation with GROUP BY and HAVING — mirroring the workflow of a professional data analyst.

The dataset analysis revealed a business generating over $1.26 million in total order value, with a healthy average order value of $1,053.97. However, the high cancellation and return rate (41.4% combined) represents the most critical area for business improvement. From a marketing perspective, Instagram is the dominant acquisition channel and should continue to receive prioritized budget allocation.

The SQL skills demonstrated in this project form the foundational layer of all data analytics work — enabling analysts to bridge the gap between raw data and executive-level business decisions with precision, speed, and reproducibility.

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